Exercise 01
SQL Foundations

Selecting & Sorting

(01) Retrive all Columns from the employee table.

```
SELECT *
FROM EMPLOYEES;
```

Table Employees.

ID	first_name	last_name	department	Salary	Hire	City
1	Alice	Green	IT	70000	2020-01/0	Johannesburg
2	Brian	Lee	HR	45000	2020/03/20	Cape Town
3	Cathy	Zulu	Finance	65000	2018/07/15	Durban
4	David	Mokoena	Marketing	50000	2021/11/06	pretoria
5	Eva	Naidoo	IT	72000	2017/09/25	Johannesburg

(02) Find all unique departments.

```
SELECT DISTINCT DEPARTMENT
FROM EMPLOYEES;
```

Department
IT
HR
Finance
Marketing

(03) Retrive First and last name order by Salary

```
SELECT First_name, last_Name
FROM EMPOYEES
ORDER BY salary DESC;
```

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First_name	Last_Name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

(04) Retrive the top 3 highest-paid employees.

```
SELECT ID, First_name, Last_name, Salary
FROM EMPLOYEES
ORDER BY Salary DESC
LIMIT 3;
```

ID	First_name	Last_name	Salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

Filtering ~~&~~ with Where

(05) Find employees in the IT department.

```
SELECT id, first_name, last_name, department
FROM EMPLOYEES
WHERE DEPARTMENT='IT';
```

ID	First_name	Last_name	Department
1	Alice	Green	IT
5	Eva	Naidoo	IT

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(06) Find employees in Finance with Salary > 60000

```
SELECT Id, First_name, last_name, department, Salary
FROM EMPLOYEES
Where department = 'Finance' and salary > 60000;
```

ID	First_name	last_name	Department	Salary
3	Cathy	Zulu	Finance	65000

(07) Find employees in HR or Marketing

```
SELECT Id, First_name, last_name, department
FROM EMPLOYEES
WHERE DEPARTMENT = 'HR' and DEPARTMENT
= 'MARKETING';
```

ID	First_name	last_name	Department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

(08) Find employees not in IT.

```
SELECT Id, First_name, last_name, department
FROM EMPLOYEES
WHERE Department <> 'IT';
```

ID	First_name	last_name	Department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

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(9) Find Employees in IT, HR or Finance

```
SELECT Id, last_name, First_name, department
FROM EMPLOYEES
Where Department IN ('IT', 'HR', 'Finance');
```

ID	First_name	last_name	Department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

(10) Find employees in IT with Salary > 65000 and City Johannesburg.

```
SELECT Id,
First_name,
Last_name,
Department,
Salary,
City
FROM EMPLOYEES
WHERE department = 'IT' And Salary > 65000
and City = 'Johannesburg';
```

ID	First-name	last-name	Department	Salary	City
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Naidoo	IT	72000	Johannesburg

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